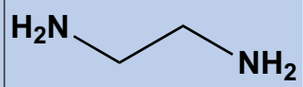
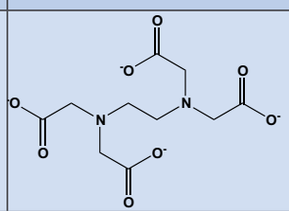
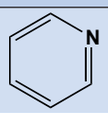
Common name	Formula	IUPAC ligand name
Bromide	Br^-	bromido
Chloride	Cl^-	chlorido
Fluoride	F^-	fluorido
Cyanide	CN^-	cyanido
Hydroxide	OH^-	hydroxido
Carbonate	CO_3^{2-}	carbonato
Nitrate	NO_3^-	nitrato
Nitrite	NO_2^-	$\leftarrow NO_2^-$; nitrito-κN $\leftarrow ONO^-$; nitrito-κO
Sulphate	SO_4^{2-}	sulphato
Sulphide	S^{2-}	sulphido
Oxalate (ox)	$C_2O_4^{2-}$	oxalato
Ethylenediamine (en)		ethane-1,2-diamine
Ethylenediaminetetraacetate (EDTA)		2,2',2'',2'''-(ethane-1,2-diyldinitrilo)tetraacetato
Triphenylphosphine	$P(Ph)_3$	triphenylphosphane
Pyridine (py)		pyridine

iii. If the coordination entity contains more than one ligand of a particular type, the multiples of ligand (2, 3, 4 etc...) is indicated by adding appropriate Greek prefixes (di, tri, tetra, etc...) to the name of the ligand. If the name of a ligand itself contains a Greek prefix (eg. ethylenediamine), use an alternate prefixes (bis, tris, tetrakis etc..) to specify the multiples of such ligands. These numerical prefixes are not taken into account for alphabetising the name of ligands.

b. Naming the central metal: In cationic/neutral complexes, the element name is used as such for naming the central metal atom/ion, whereas, a suffix 'ate' is used along with the element name in anionic complexes. The oxidation state of the metal is written immediately after the metal name using roman numerals in parenthesis.

XII U5 Coordination jagan.indd 137 22-12-2022 19:51:01